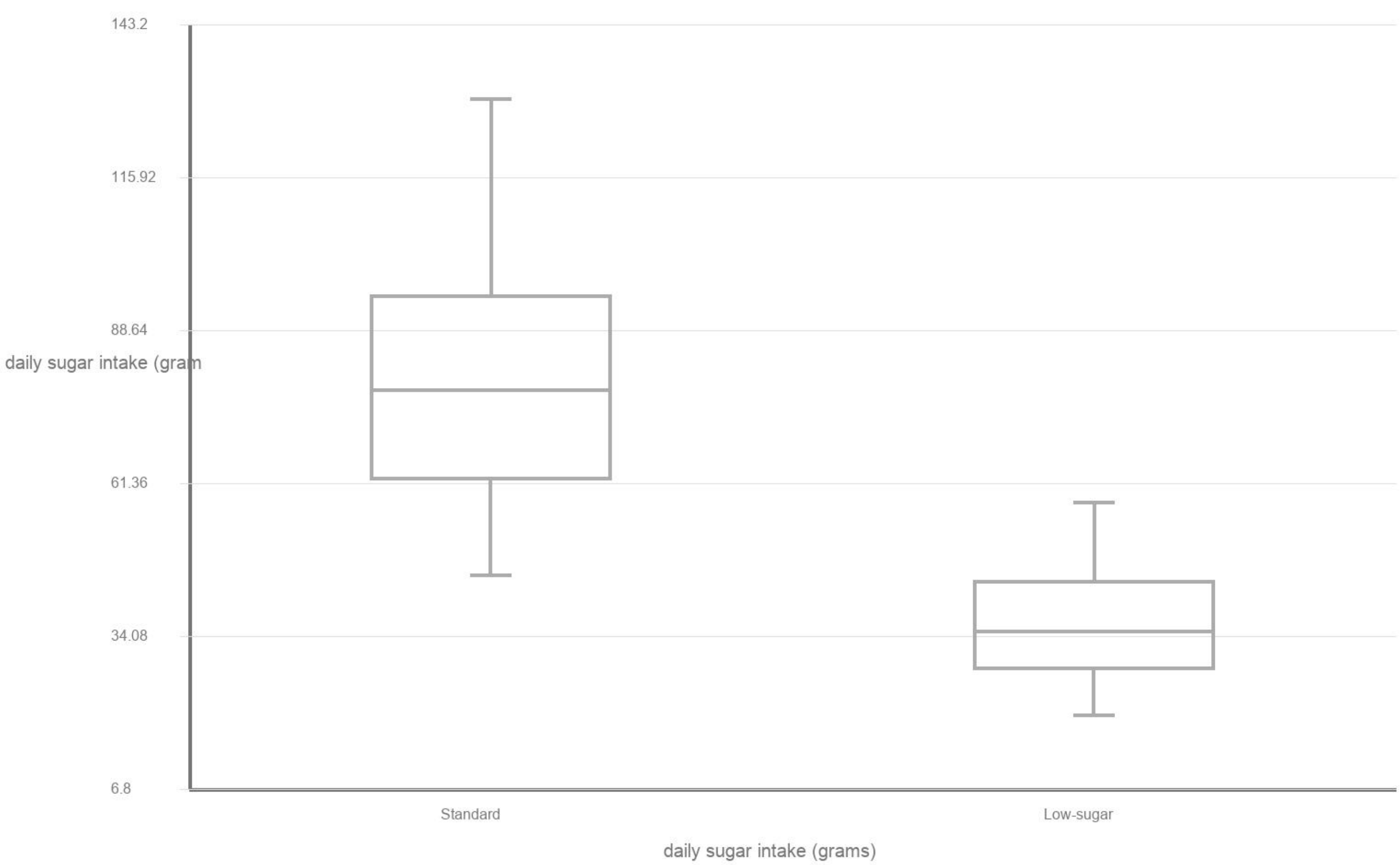
APSTATS-HDG-2026-GRAPH-013

A nutritionist compared daily sugar intake (grams) for two diet groups. Construct side-by-side boxplots from the five-number summaries, using one common scale. Then write one sentence comparing the centers and one sentence comparing the variability.

Data table

Diet group	Min	Q1	Median	Q3	Max
Standard	45	62	78	95	130
Low-sugar	20	28	35	44	58

Trace or redraw the faint graph below on paper. Keep the item ID visible on the page before photographing.



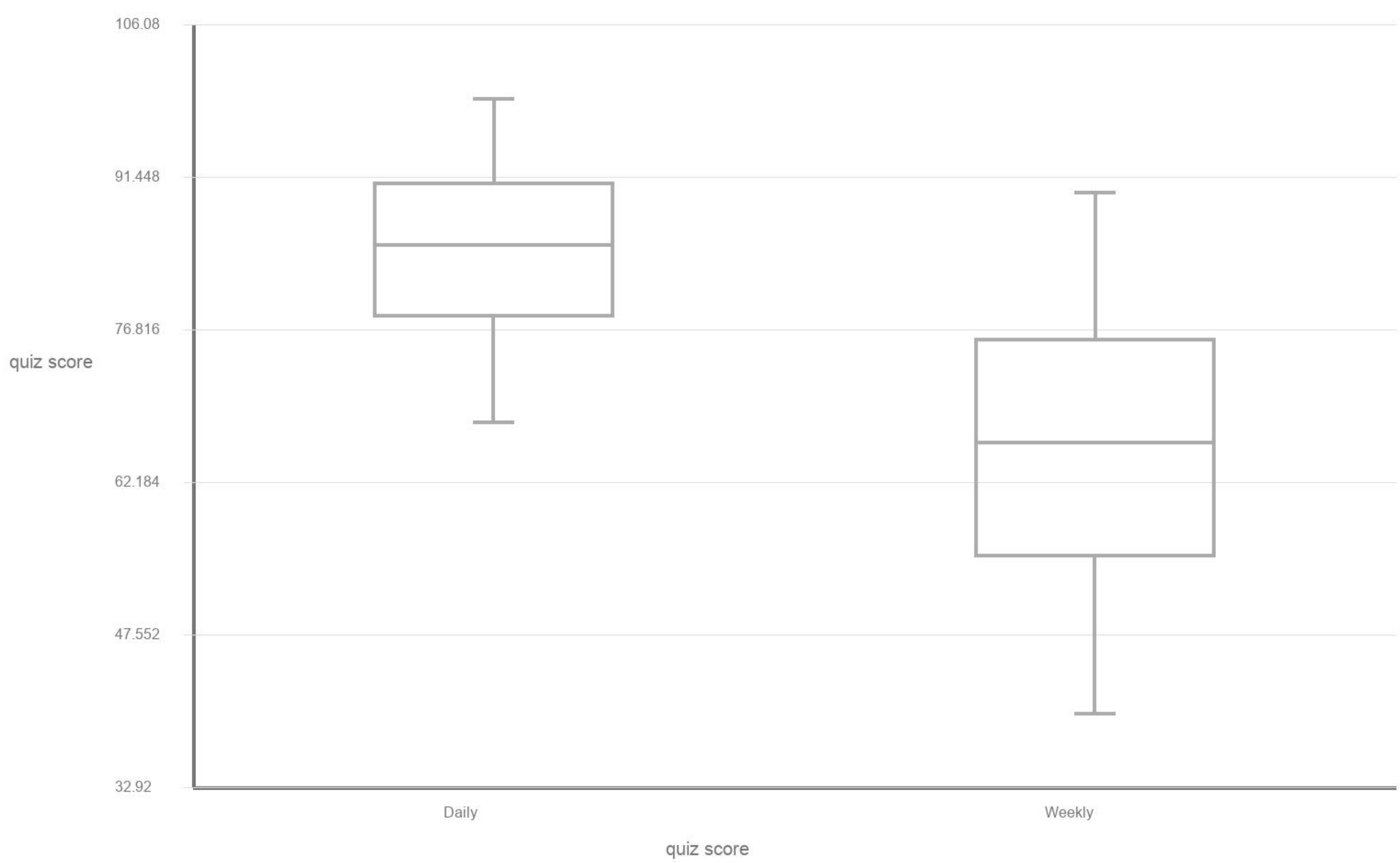
APSTATS-HDG-2026-GRAPH-014

A tutoring center compared quiz scores for students who used the app daily versus weekly. Construct side-by-side boxplots from the five-number summaries using one common scale. Then compare the typical score and the spread in context.

Data table

Usage	Min	Q1	Median	Q3	Max
Daily	68	78	85	91	99
Weekly	40	55	66	76	90

Trace or redraw the faint graph below on paper. Keep the item ID visible on the page before photographing.



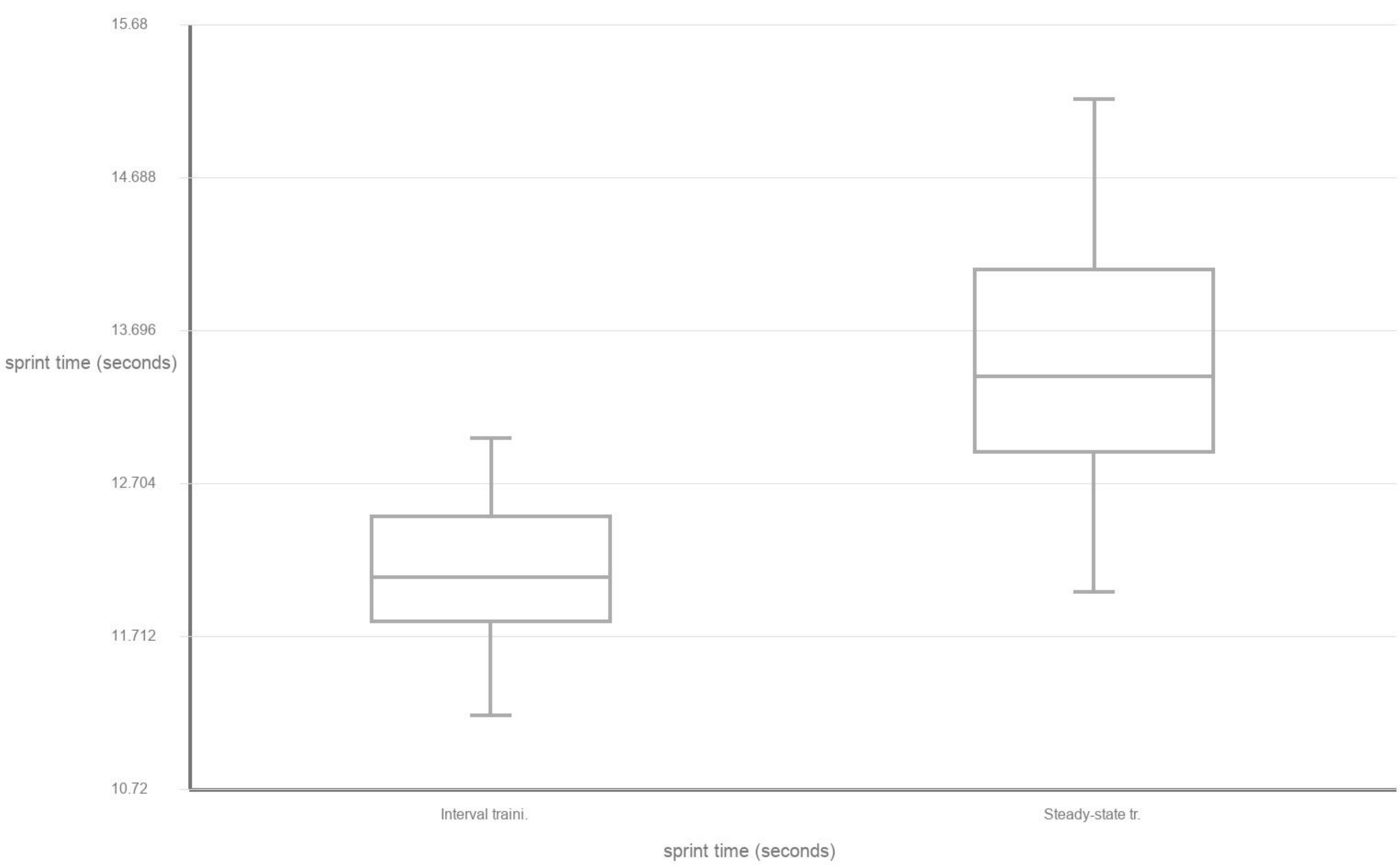
APSTATS-HDG-2026-GRAPH-015

A coach compared sprint times (seconds) for two training groups. Construct side-by-side boxplots from the five-number summaries using one common scale. Then compare the typical time and the spread in context.

Data table

Group	Min	Q1	Median	Q3	Max
Interval training	11.2	11.8	12.1	12.5	13.0
Steady-state traini.	12.0	12.9	13.4	14.1	15.2

Trace or redraw the faint graph below on paper. Keep the item ID visible on the page before photographing.



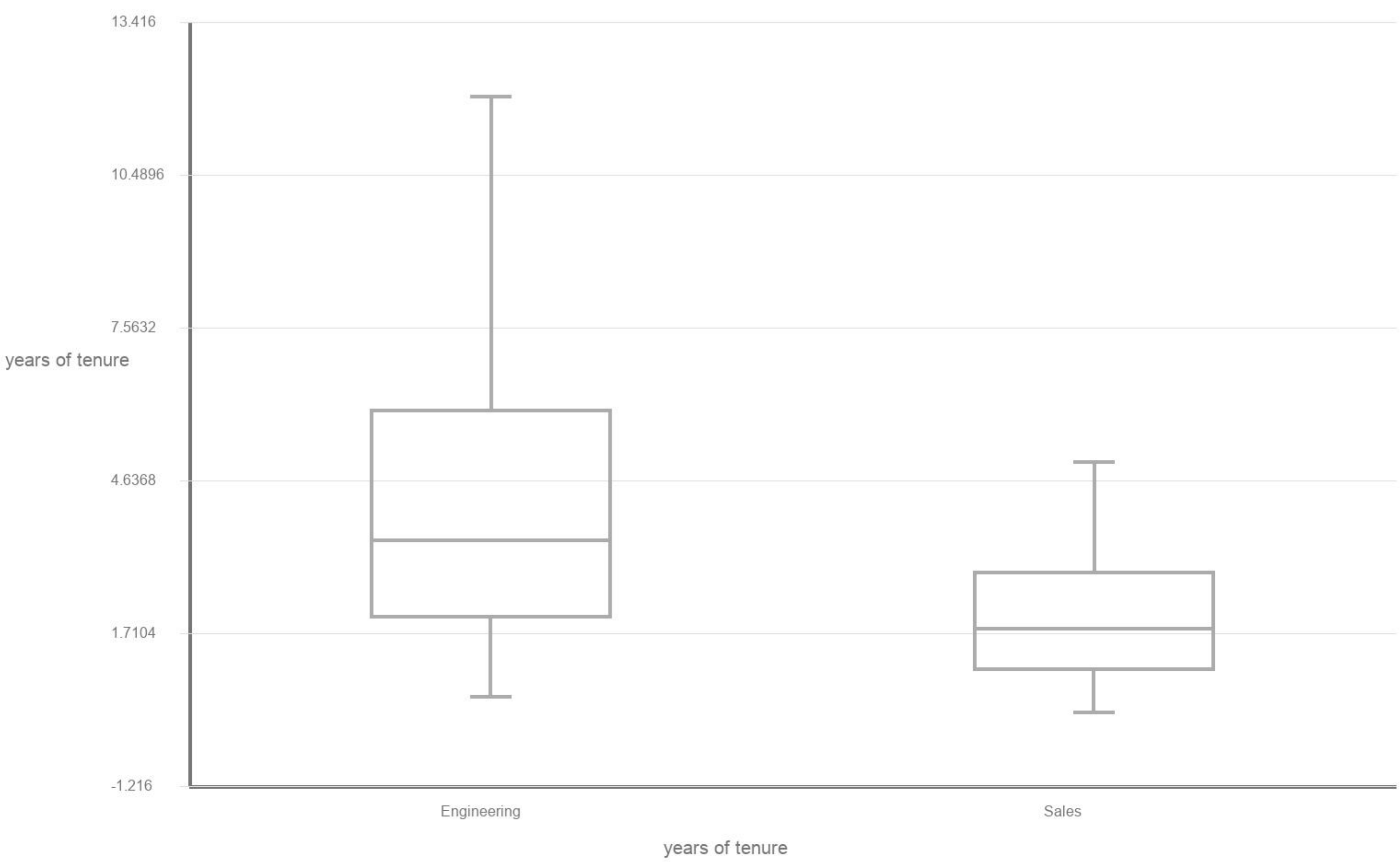
APSTATS-HDG-2026-GRAPH-016

An HR analyst compared years of tenure for two departments. Construct side-by-side boxplots from the five-number summaries using one common scale. Then compare the typical tenure and the spread in context.

Data table

Department	Min	Q1	Median	Q3	Max
Engineering	0.5	2.0	3.5	6.0	12.0
Sales	0.2	1.0	1.8	2.9	5.0

Trace or redraw the faint graph below on paper. Keep the item ID visible on the page before photographing.



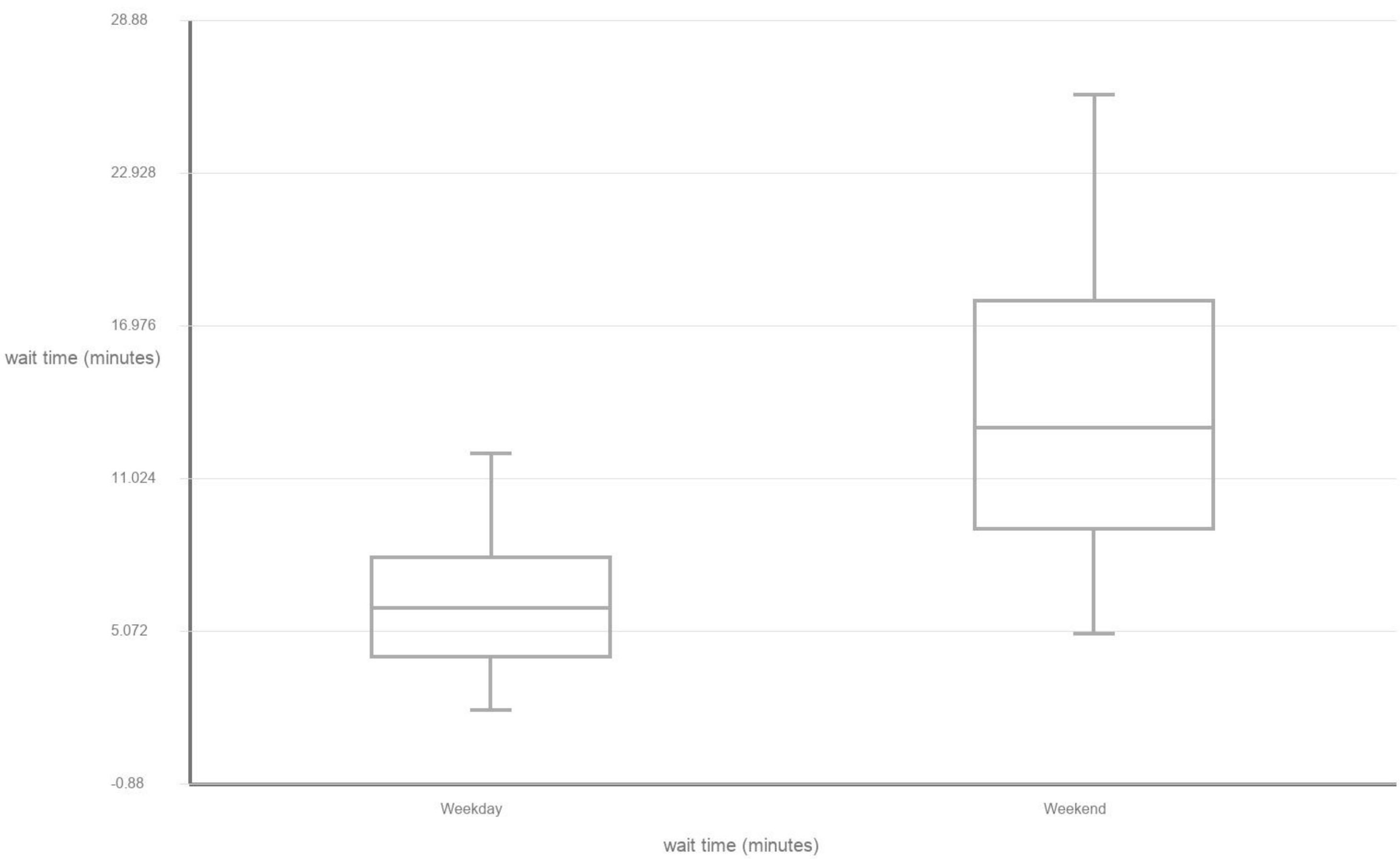
APSTATS-HDG-2026-GRAPH-017

A cafe manager compared customer wait times (minutes) on weekdays versus weekends. Construct side-by-side boxplots from the five-number summaries using one common scale. Then compare the typical wait time and the spread in context.

Data table

Day type	Min	Q1	Median	Q3	Max
Weekday	2	4	6	8	12
Weekend	5	9	13	18	26

Trace or redraw the faint graph below on paper. Keep the item ID visible on the page before photographing.



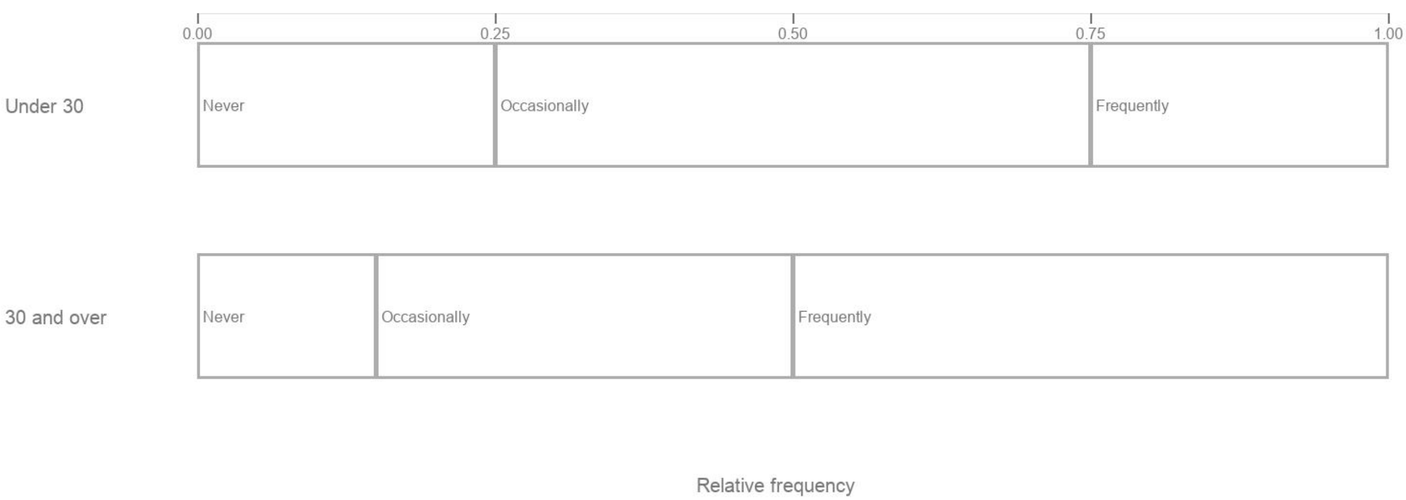
APSTATS-HDG-2026-GRAPH-018

A streaming service surveyed subscribers about how often they watch documentaries. Complete a segmented bar graph using relative frequencies for each age group. Then identify which group has the larger relative frequency of frequent viewing.

Data table

Age group	Never	Occasionally	Frequently
Under 30	25	50	25
30 and over	15	35	50

Trace or redraw the faint graph below on paper. Keep the item ID visible on the page before photographing.



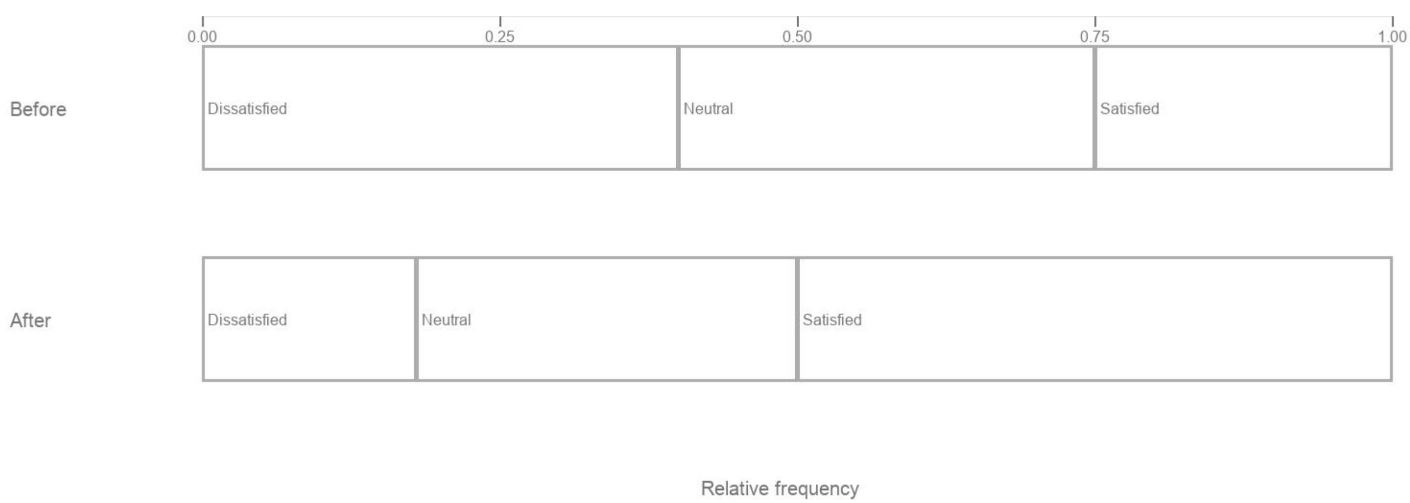
APSTATS-HDG-2026-GRAPH-019

A hospital tracked patient satisfaction ratings before and after a new scheduling system. Complete a segmented bar graph using relative frequencies for each period. Then state whether the relative frequency of 'satisfied' ratings increased.

Data table

Period	Dissatisfied	Neutral	Satisfied
Before	40	35	25
After	18	32	50

Trace or redraw the faint graph below on paper. Keep the item ID visible on the page before photographing.



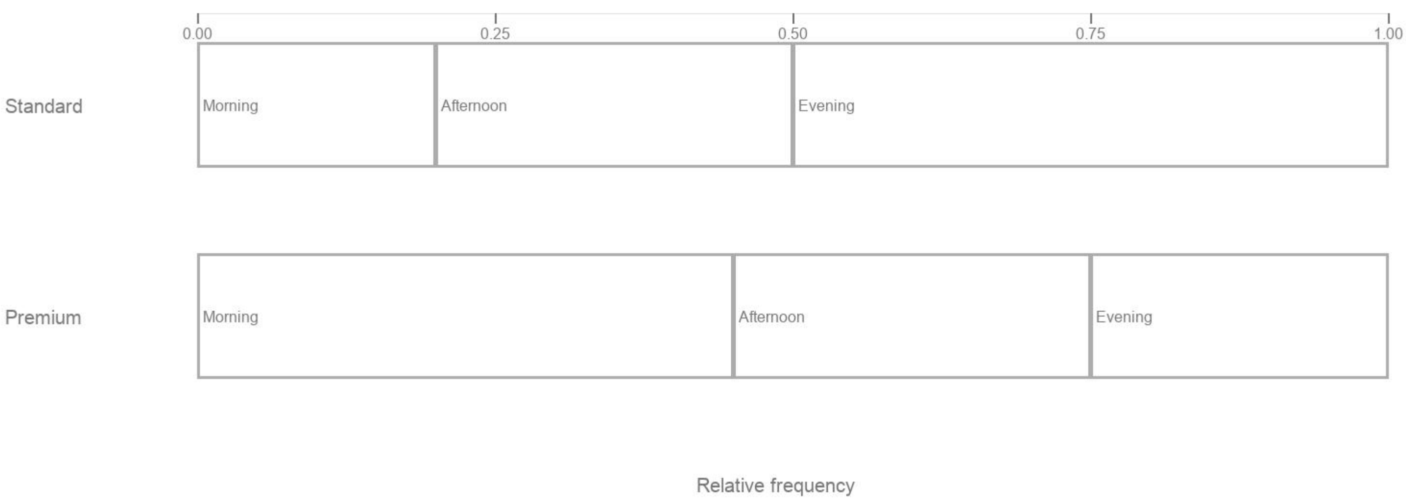
APSTATS-HDG-2026-GRAPH-020

A gym surveyed members about preferred workout time. Complete a segmented bar graph using relative frequencies for each membership type. Then identify which membership type has the larger relative frequency of morning workouts.

Data table

Membership	Morning	Afternoon	Evening
Standard	20	30	50
Premium	45	30	25

Trace or redraw the faint graph below on paper. Keep the item ID visible on the page before photographing.



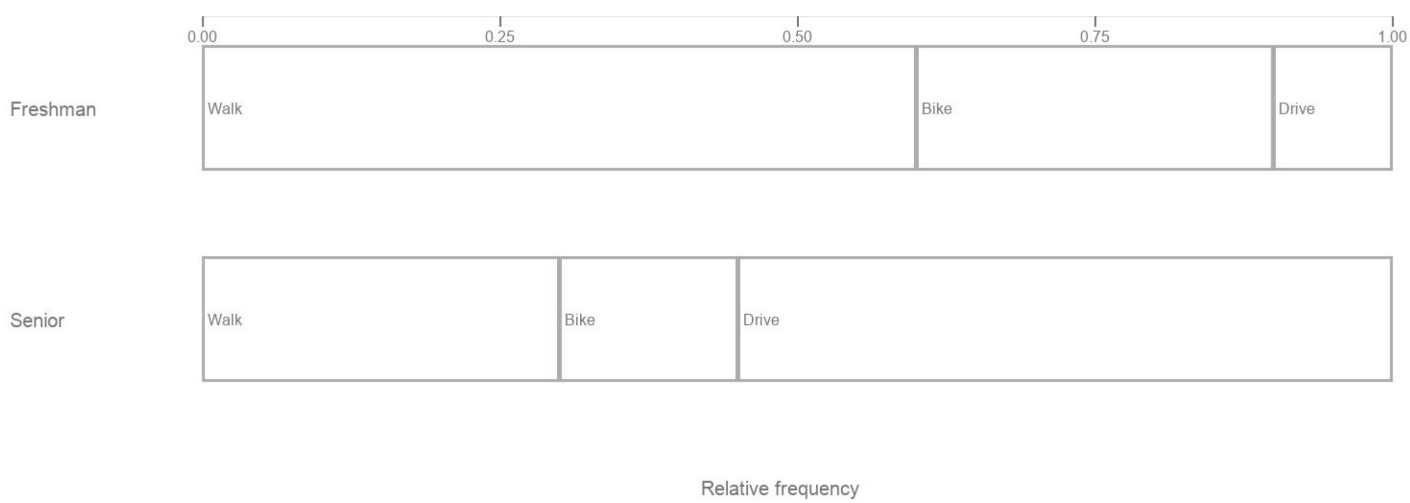
APSTATS-HDG-2026-GRAPH-021

A university surveyed students on their primary transportation mode by class year. Complete a segmented bar graph using relative frequencies for each class year. Then identify which class year has the larger relative frequency of biking.

Data table

Class year	Walk	Bike	Drive
Freshman	60	30	10
Senior	30	15	55

Trace or redraw the faint graph below on paper. Keep the item ID visible on the page before photographing.



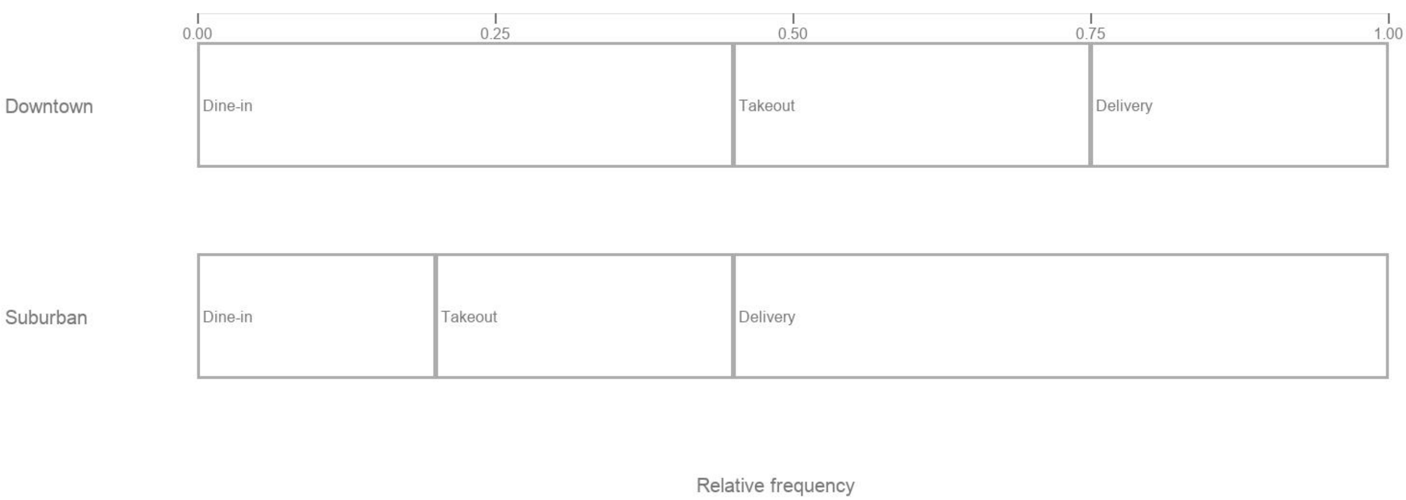
APSTATS-HDG-2026-GRAPH-022

A restaurant chain tracked order type by location. Complete a segmented bar graph using relative frequencies for each location type. Then identify which location type has the larger relative frequency of delivery orders.

Data table

Location type	Dine-in	Takeout	Delivery
Downtown	45	30	25
Suburban	20	25	55

Trace or redraw the faint graph below on paper. Keep the item ID visible on the page before photographing.



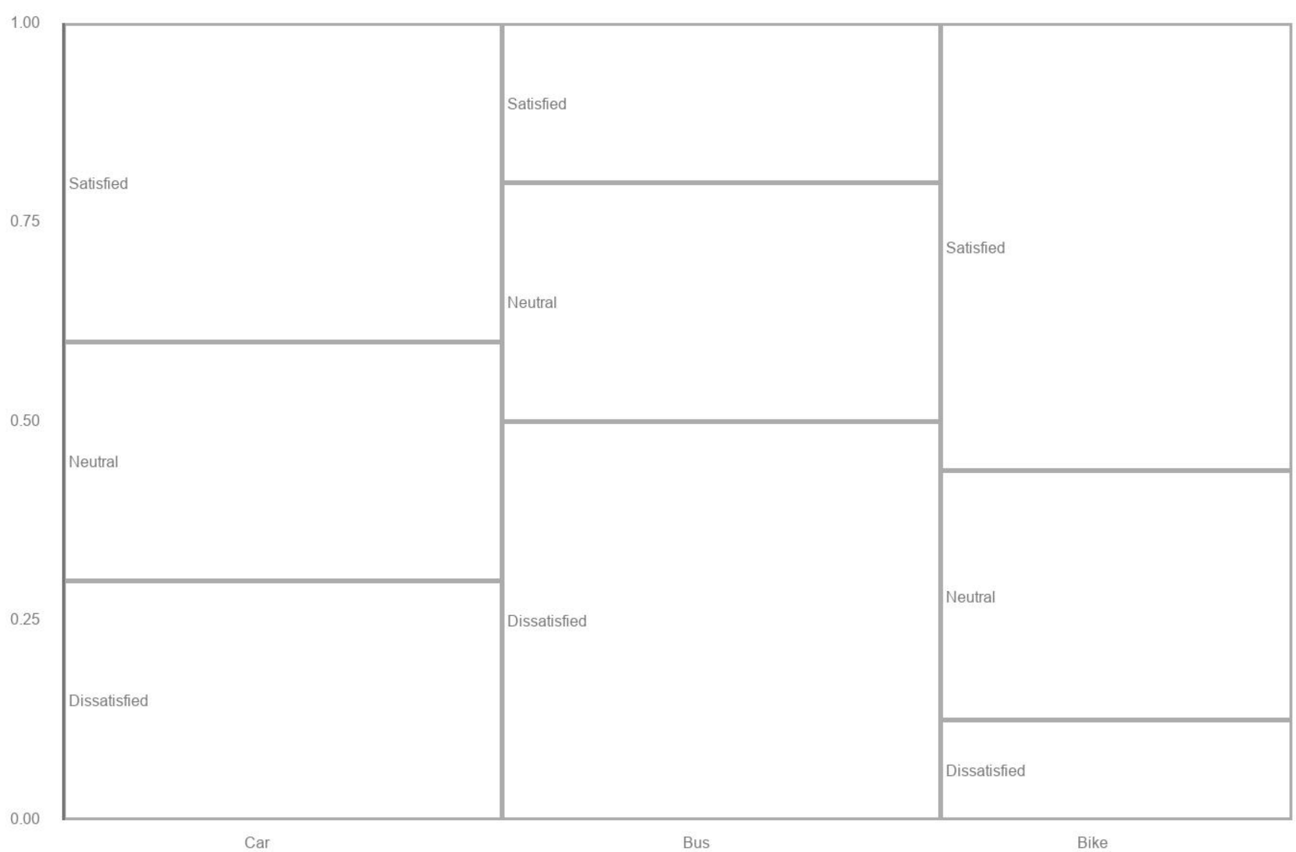
APSTATS-HDG-2026-GRAPH-023

Students were classified by primary transportation mode and commute-time satisfaction. Draw a mosaic plot from the table so that each mode's width is proportional to its total count and each vertical segment shows the conditional distribution of satisfaction within that mode. Then state which mode has the largest number of dissatisfied students.

Data table

Mode	Satisfied	Neutral	Dissatisfied
Car	40	30	30
Bus	20	30	50
Bike	45	25	10

Trace or redraw the faint graph below on paper. Keep the item ID visible on the page before photographing.



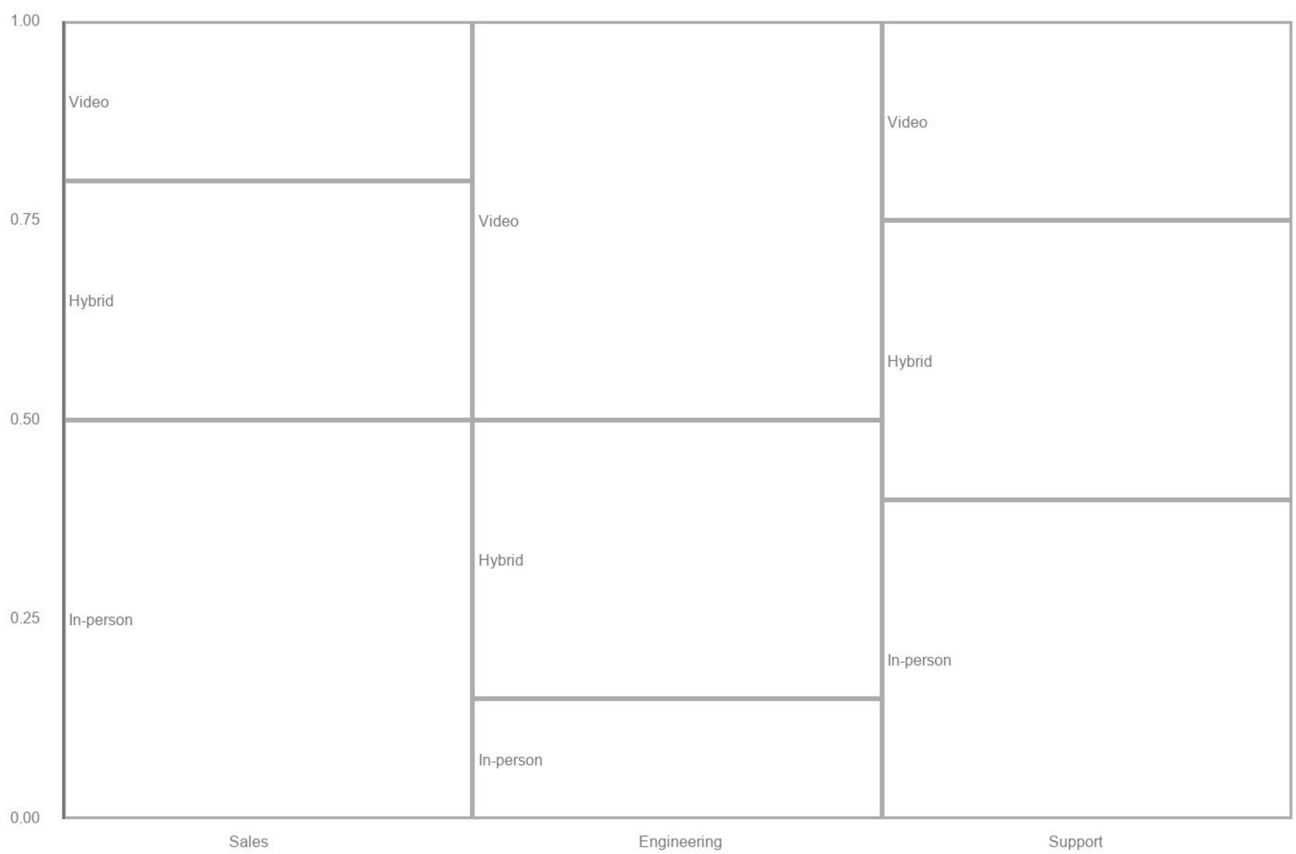
APSTATS-HDG-2026-GRAPH-024

Employees were classified by department and preferred meeting format. Draw a mosaic plot so each department width is proportional to its total count and each vertical segment shows the conditional distribution of format preference within that department. Then identify which department has the largest number of employees preferring in-person meetings.

Data table

Department	Video	Hybrid	In-person
Sales	20	30	50
Engineering	50	35	15
Support	25	35	40

Trace or redraw the faint graph below on paper. Keep the item ID visible on the page before photographing.



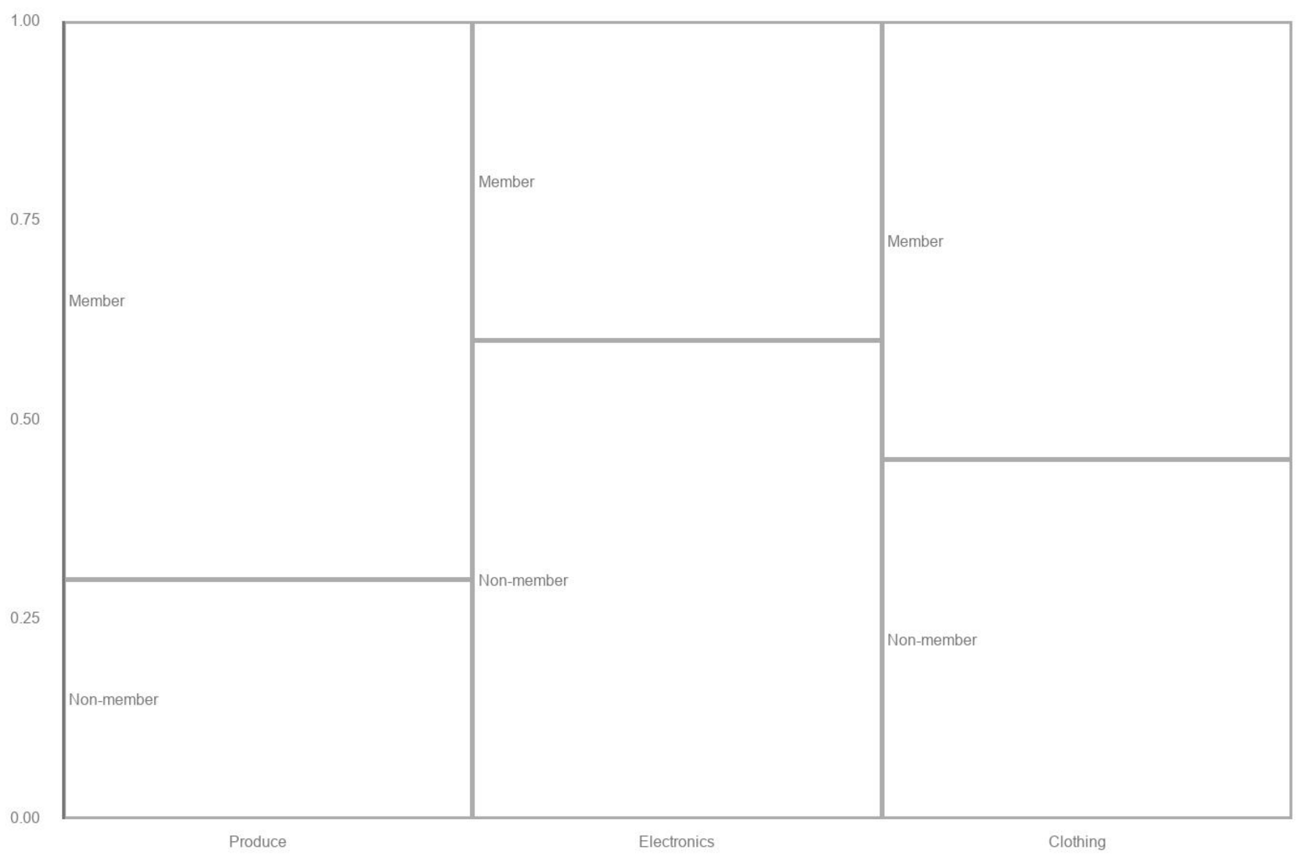
APSTATS-HDG-2026-GRAPH-025

Shoppers were classified by store section visited most and membership status. Draw a mosaic plot so each section's width is proportional to its total count and each vertical segment shows the conditional distribution of membership status within that section. Then identify which section has the largest number of member shoppers.

Data table

Section	Member	Non-member
Produce	70	30
Electronics	40	60
Clothing	55	45

Trace or redraw the faint graph below on paper. Keep the item ID visible on the page before photographing.



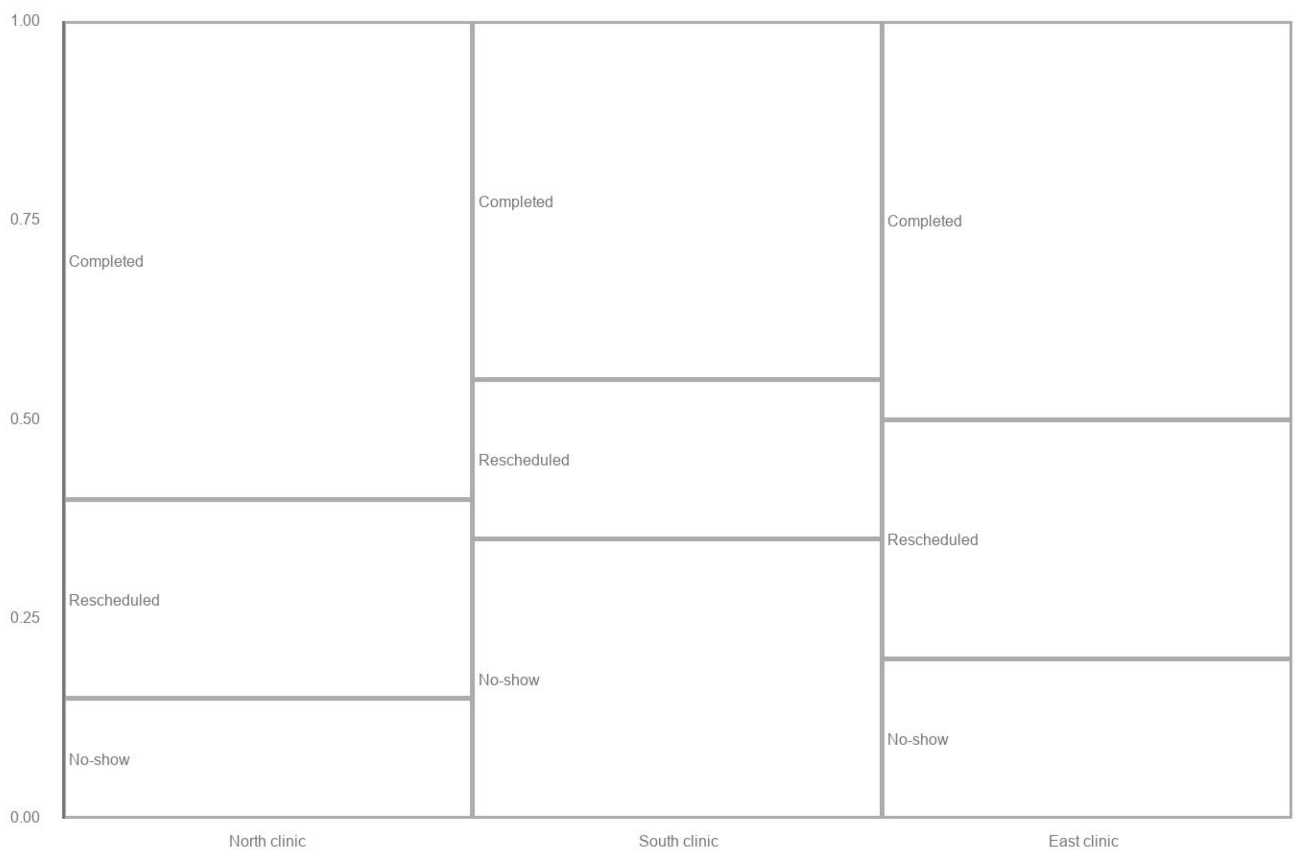
APSTATS-HDG-2026-GRAPH-026

Patients were classified by clinic location and appointment outcome. Draw a mosaic plot so each location's width is proportional to its total count and each vertical segment shows the conditional distribution of outcome within that location. Then identify which location has the largest number of no-show appointments.

Data table

Location	Completed	Rescheduled	No-show
North clinic	60	25	15
South clinic	45	20	35
East clinic	50	30	20

Trace or redraw the faint graph below on paper. Keep the item ID visible on the page before photographing.



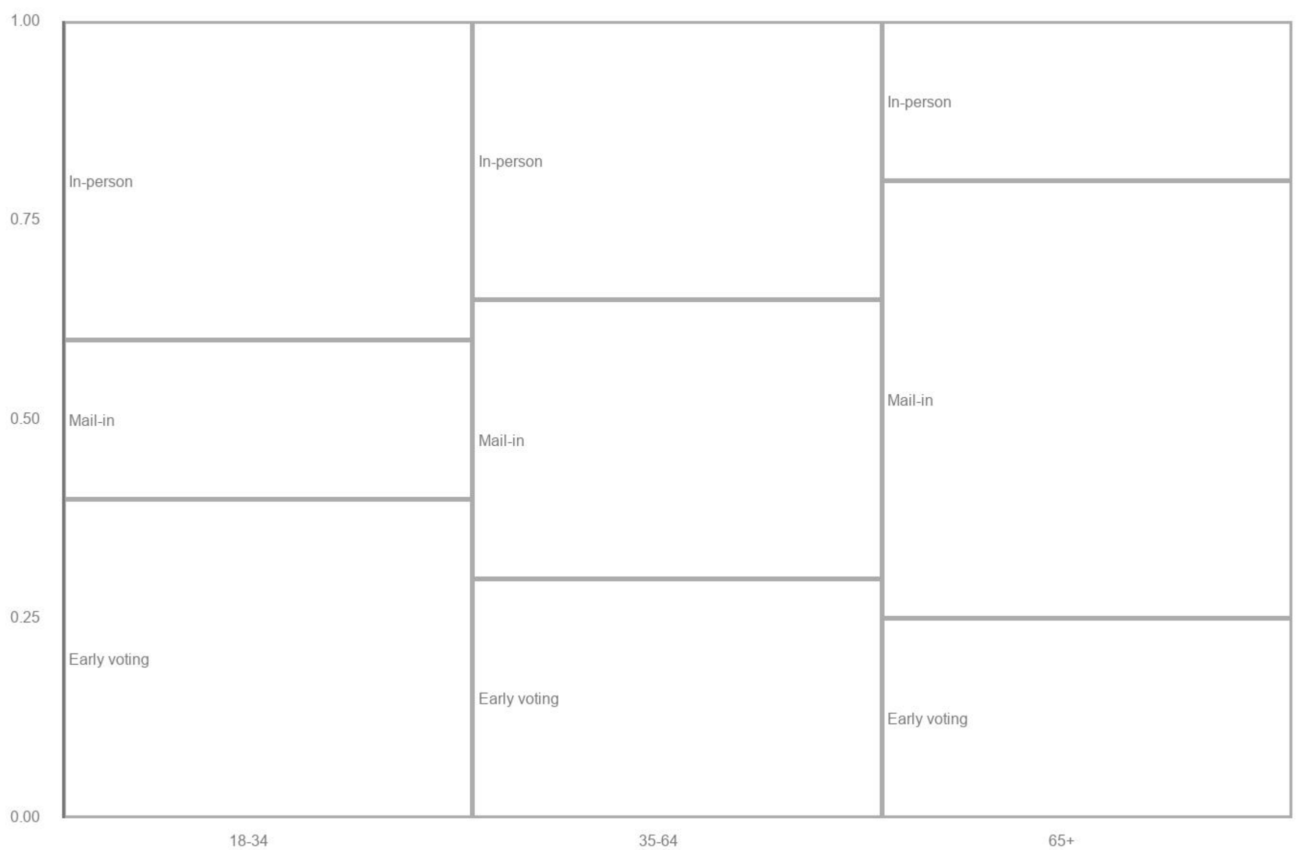
APSTATS-HDG-2026-GRAPH-027

Voters were classified by age bracket and preferred voting method. Draw a mosaic plot so each age bracket's width is proportional to its total count and each vertical segment shows the conditional distribution of voting method within that bracket. Then identify which age bracket has the largest number of mail-in voters.

Data table

Age bracket	In-person	Mail-in	Early voting
18-34	40	20	40
35-64	35	35	30
65+	20	55	25

Trace or redraw the faint graph below on paper. Keep the item ID visible on the page before photographing.



APSTATS-HDG-2026-GRAPH-028

A sample of 12 commute times (minutes) is shown below. Construct a dotplot of the data, then describe the distribution's shape and identify any possible outlier.

Data table

Commute time (min)
15
18
18
19
20
20
20
21
22
22
24
38

Trace or redraw the faint graph below on paper. Keep the item ID visible on the page before photographing.

APSTATS-HDG-2026-GRAPH-029

A sample of 12 test scores is shown below. Construct a dotplot of the data, then describe the distribution's shape and comment on whether there is a clear outlier.

Data table

Test score
62
78
80
81
82
83
84
84
85
86
87
88

Trace or redraw the faint graph below on paper. Keep the item ID visible on the page before photographing.

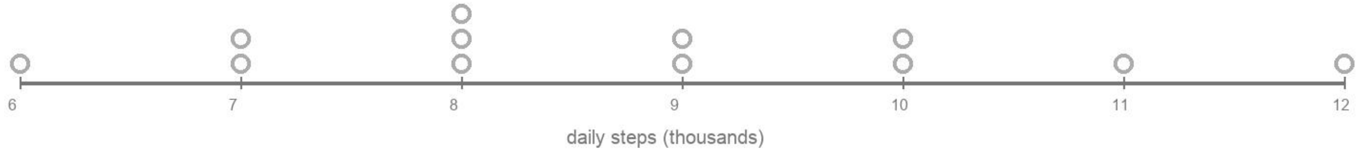
APSTATS-HDG-2026-GRAPH-030

A sample of 12 daily step counts (thousands) is shown below. Construct a dotplot of the data, then describe the distribution's shape and comment on whether there is a clear outlier.

Data table

Steps (thousands)
6
7
7
8
8
8
9
9
10
10
11
12

Trace or redraw the faint graph below on paper. Keep the item ID visible on the page before photographing.



APSTATS-HDG-2026-GRAPH-031

A sample of 12 package delivery times (days) is shown below. Construct a dotplot of the data, then describe the distribution's shape and identify any possible outlier.

Data table

Delivery time (days)
1
2
2
2
3
3
3
3
4
4
5
11

Trace or redraw the faint graph below on paper. Keep the item ID visible on the page before photographing.

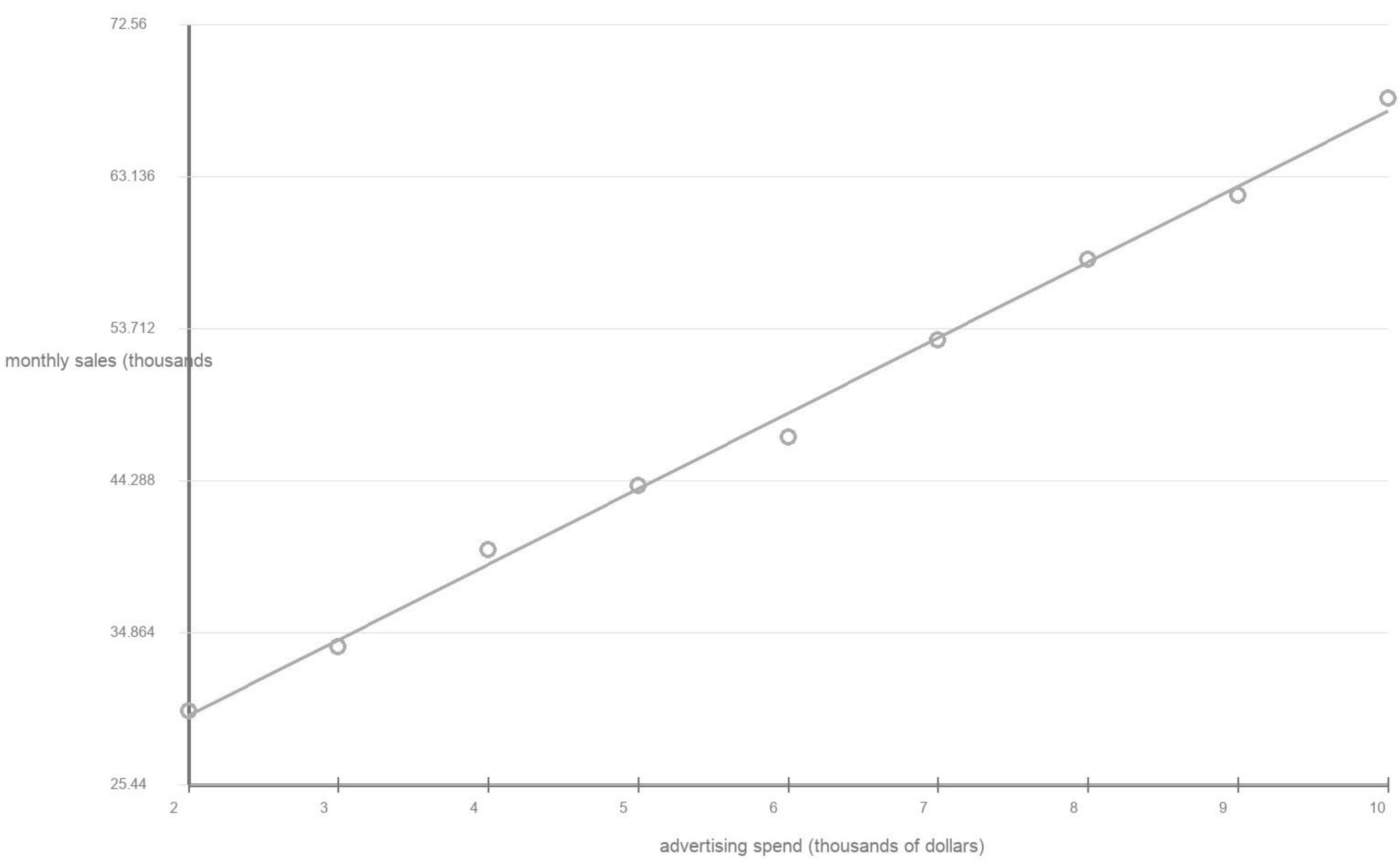
APSTATS-HDG-2026-GRAPH-032

A researcher recorded advertising spend (thousands of dollars) and monthly sales (thousands of dollars) for nine stores. Construct a scatterplot, sketch a reasonable least-squares trend line, and describe the direction, form, and strength of the association in context.

Data table

Ad spend	Sales
2	30
3	34
4	40
5	44
6	47
7	53
8	58
9	62
10	68

Trace or redraw the faint graph below on paper. Keep the item ID visible on the page before photographing.



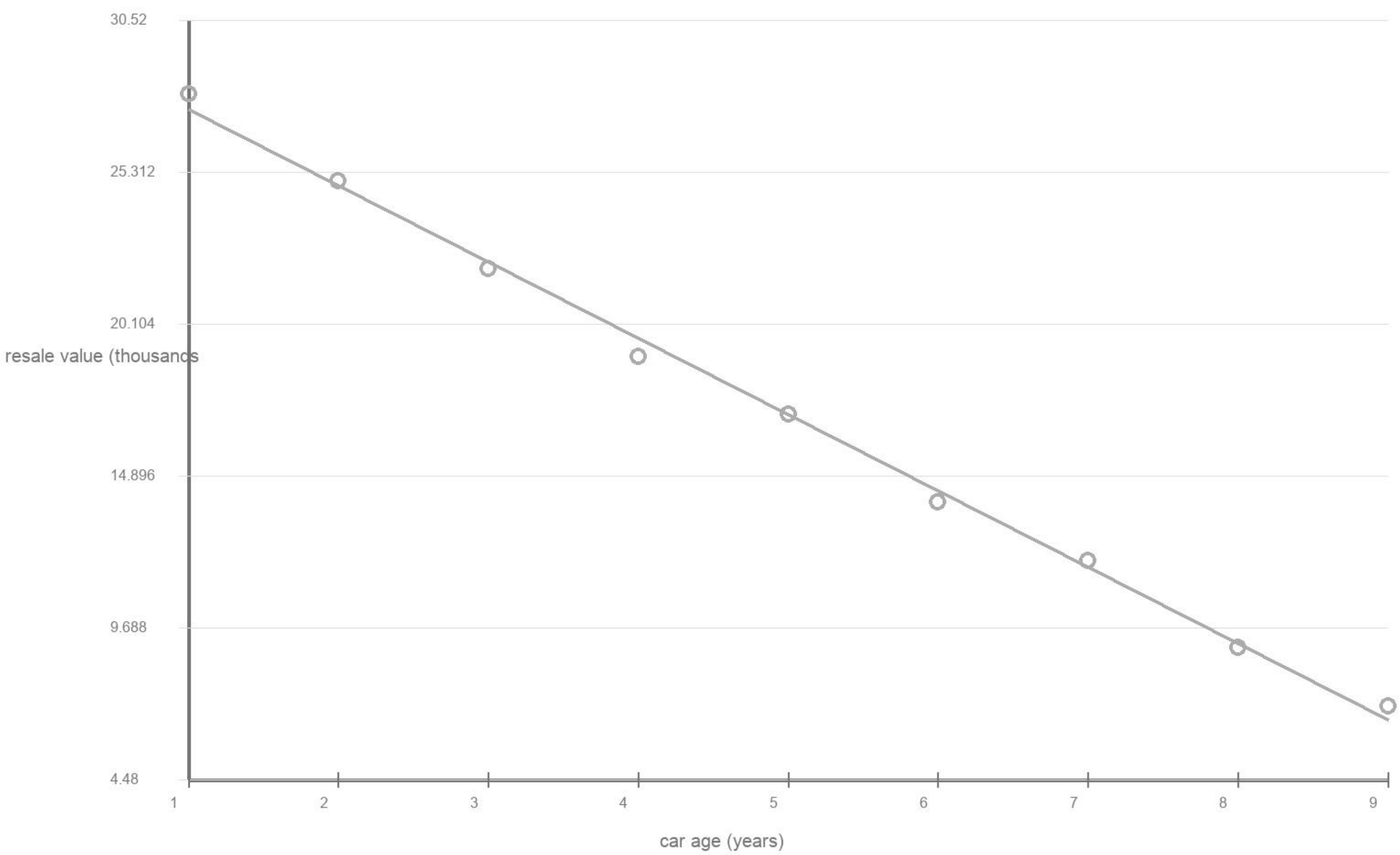
APSTATS-HDG-2026-GRAPH-033

A mechanic recorded a car's age (years) and its resale value (thousands of dollars) for nine cars. Construct a scatterplot, sketch a reasonable least-squares trend line, and describe the direction, form, and strength of the association in context.

Data table

Age	Resale value
1	28
2	25
3	22
4	19
5	17
6	14
7	12
8	9
9	7

Trace or redraw the faint graph below on paper. Keep the item ID visible on the page before photographing.



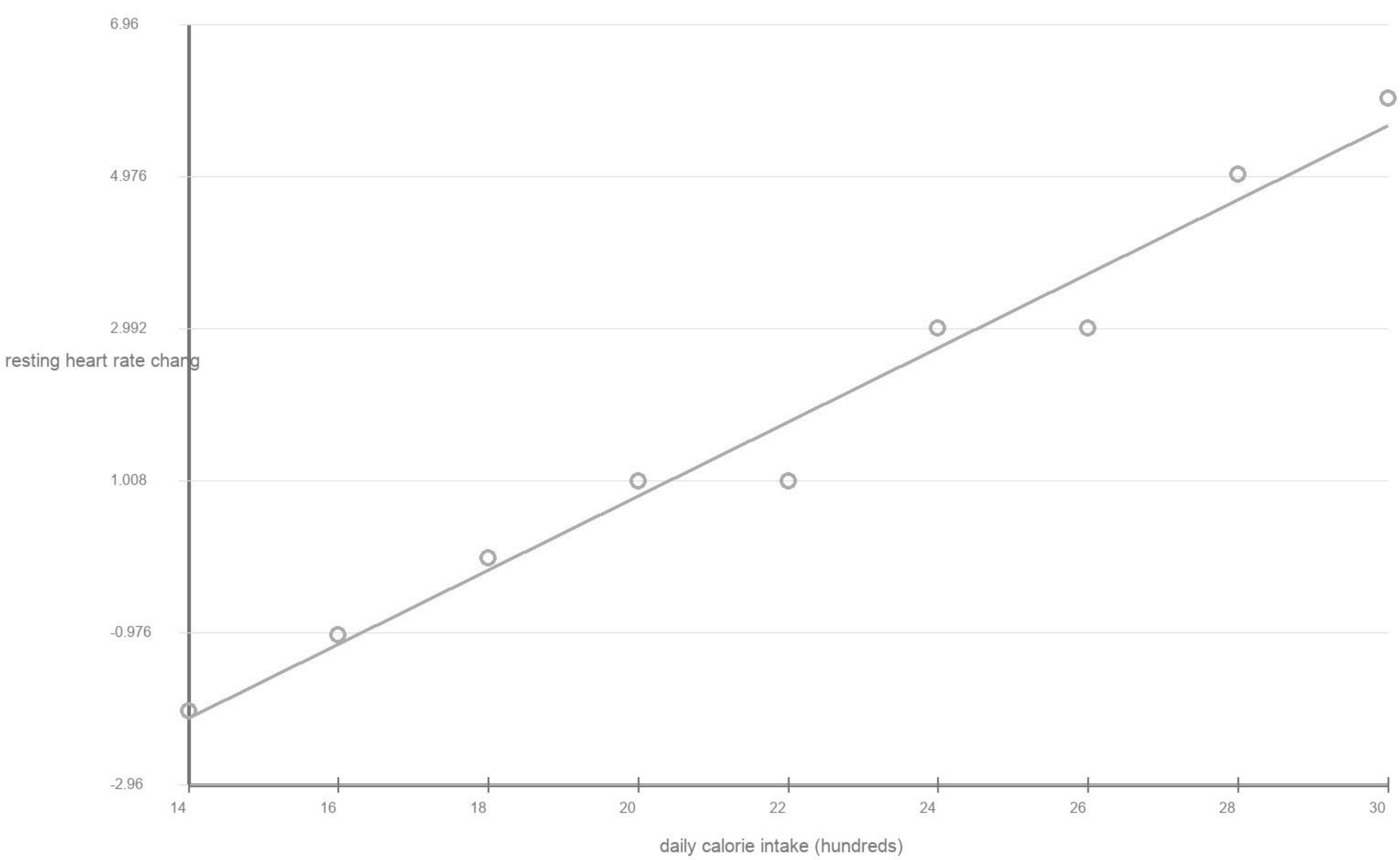
APSTATS-HDG-2026-GRAPH-034

A nutritionist recorded daily calorie intake (hundreds) and resting heart rate change (bpm) for nine participants in a controlled diet study. Construct a scatterplot, sketch a reasonable least-squares trend line, and describe the direction, form, and strength of the association in context.

Data table

Calorie intake (hundr.	Heart rate change (bp.
14	-2
16	-1
18	0
20	1
22	1
24	3
26	3
28	5
30	6

Trace or redraw the faint graph below on paper. Keep the item ID visible on the page before photographing.



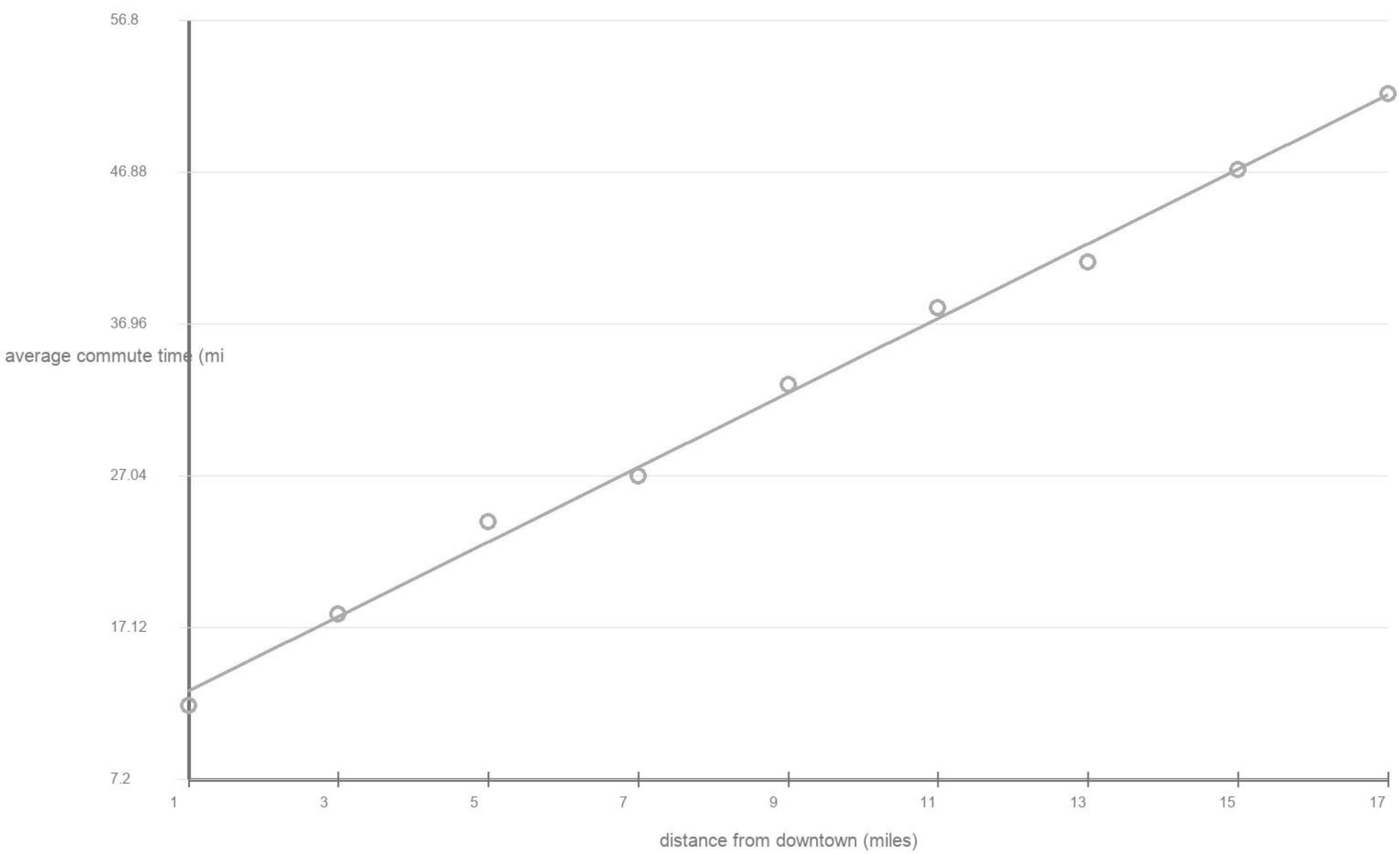
APSTATS-HDG-2026-GRAPH-035

A city planner recorded a neighborhood's distance from downtown (miles) and average commute time (minutes) for nine neighborhoods. Construct a scatterplot, sketch a reasonable least-squares trend line, and describe the direction, form, and strength of the association in context.

Data table

Distance (miles)	Commute time (min)
1	12
3	18
5	24
7	27
9	33
11	38
13	41
15	47
17	52

Trace or redraw the faint graph below on paper. Keep the item ID visible on the page before photographing.



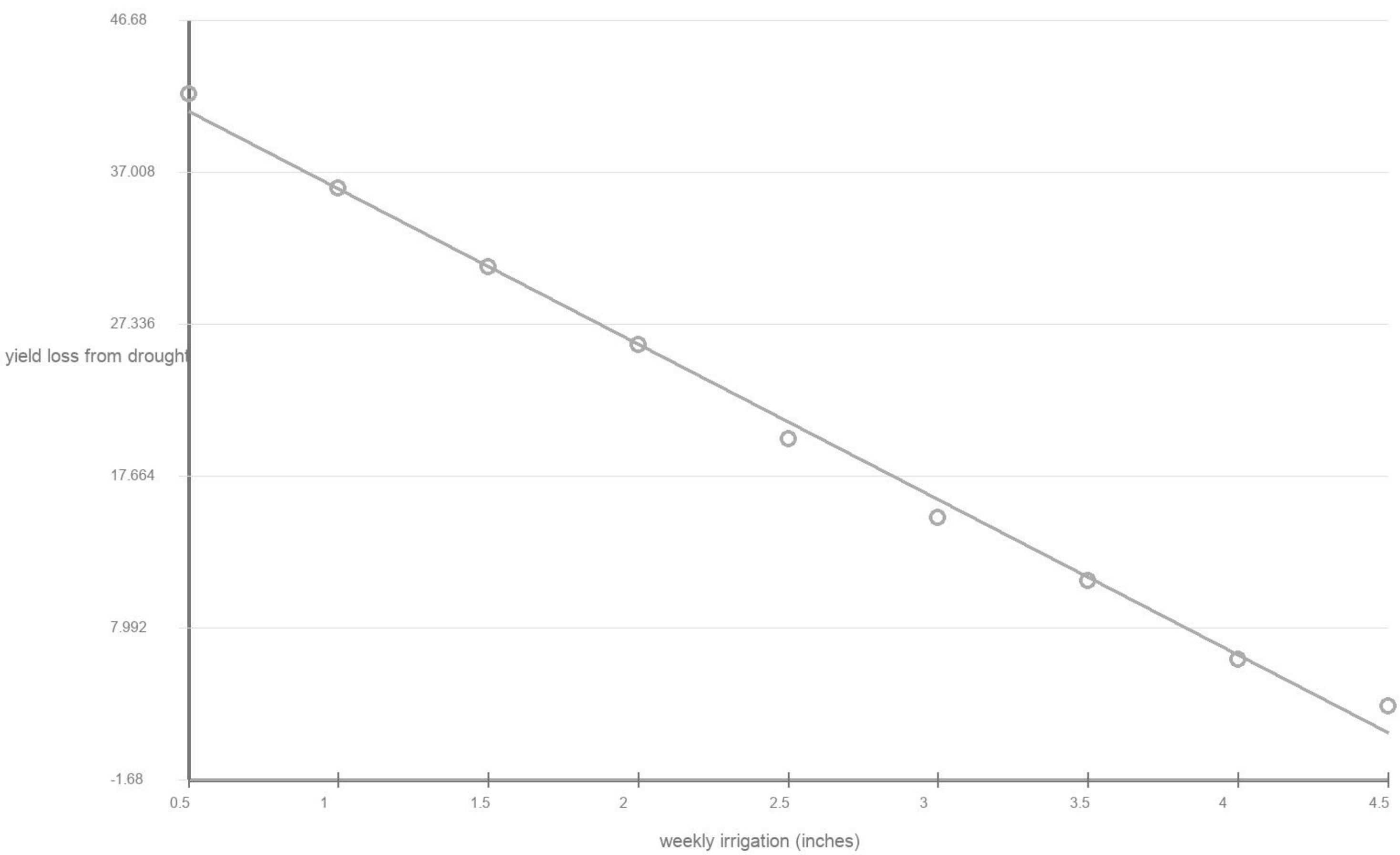
APSTATS-HDG-2026-GRAPH-036

A farmer recorded weekly irrigation amount (inches) and crop yield loss due to drought stress (percent) for nine plots, where more irrigation is associated with less drought stress. Construct a scatterplot, sketch a reasonable least-squares trend line, and describe the direction, form, and strength of the association in context.

Data table

Irrigation (in)	Yield loss (%)
0.5	42
1.0	36
1.5	31
2.0	26
2.5	20
3.0	15
3.5	11
4.0	6
4.5	3

Trace or redraw the faint graph below on paper. Keep the item ID visible on the page before photographing.



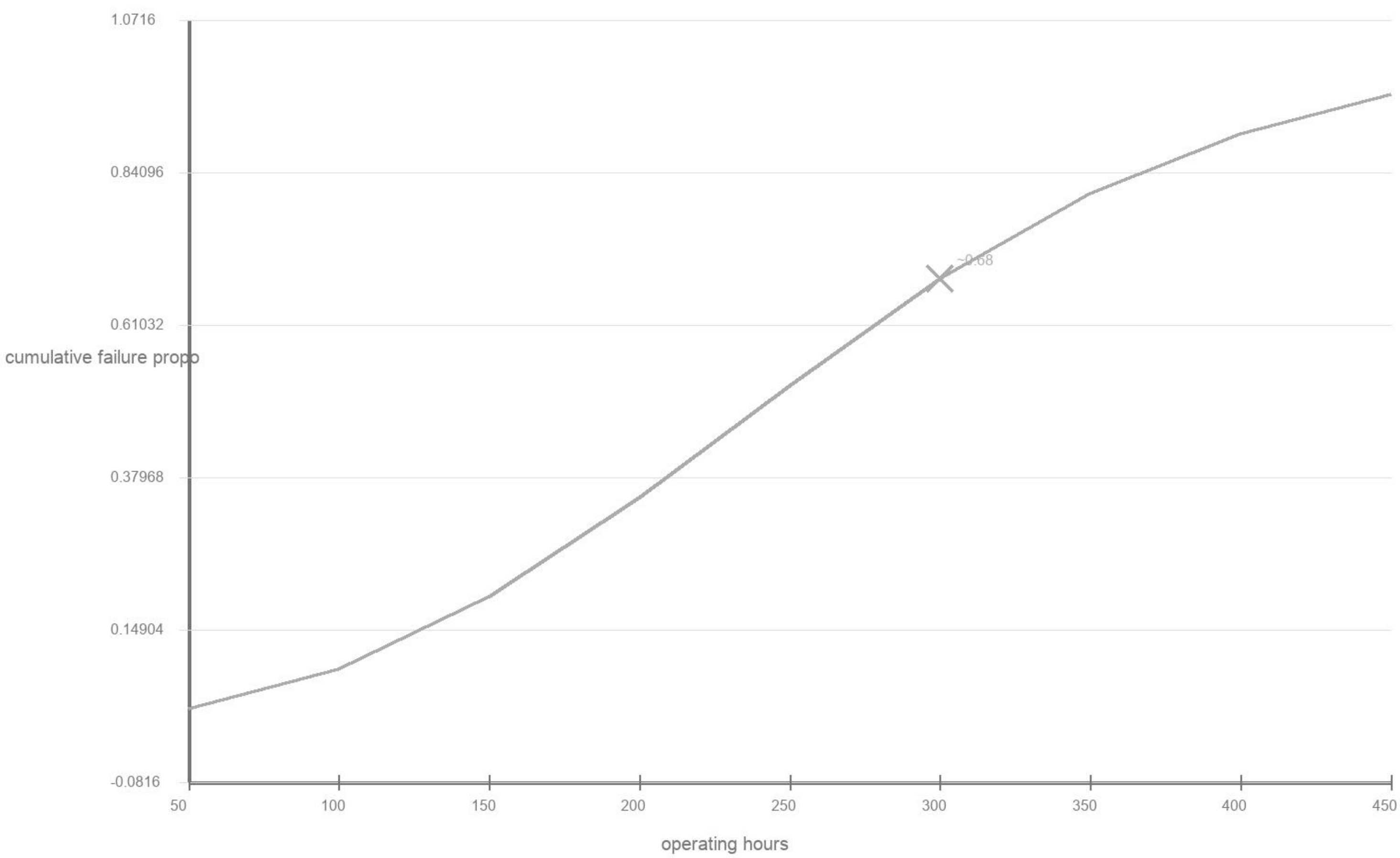
APSTATS-HDG-2026-GRAPH-037

The table gives values from a curve showing the cumulative proportion of a batch of components that have failed as operating hours increase. Draw the curve on a graph with operating hours on the x-axis and cumulative failure proportion on the y-axis. Mark the point for 300 hours with an X, then estimate the corresponding failure proportion.

Data table

Hours	Failure proportion
50	0.03
100	0.09
150	0.2
200	0.35
250	0.52
300	0.68
350	0.81
400	0.9
450	0.96

Trace or redraw the faint graph below on paper. Keep the item ID visible on the page before photographing.



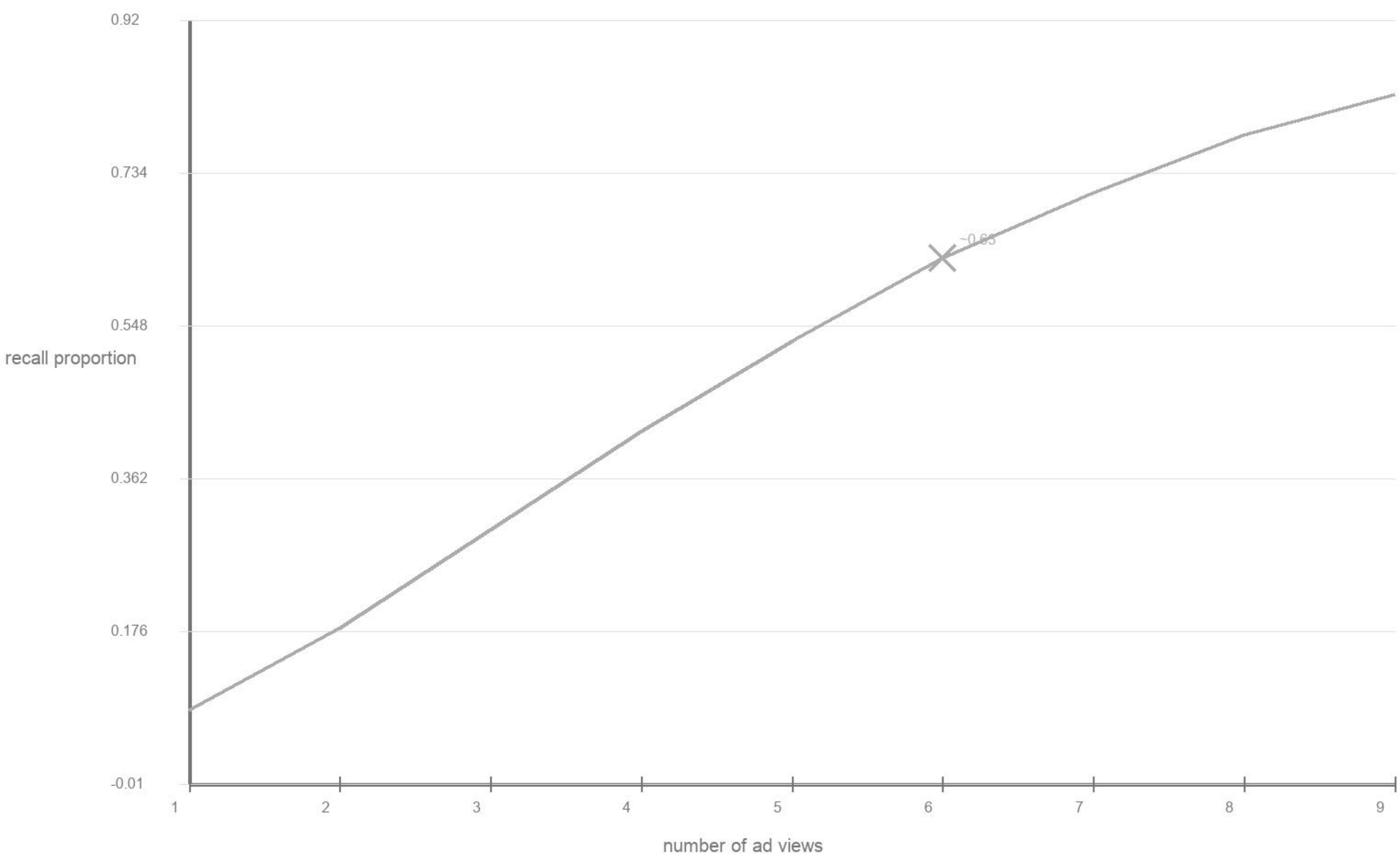
APSTATS-HDG-2026-GRAPH-038

The table gives values from a curve showing the proportion of survey respondents who recall a product name as advertising exposure (number of ad views) increases. Draw the curve on a graph with ad views on the x-axis and recall proportion on the y-axis. Mark the point for 6 ad views with an X, then estimate the corresponding recall proportion.

Data table

Ad views	Recall proportion
1	0.08
2	0.18
3	0.3
4	0.42
5	0.53
6	0.63
7	0.71
8	0.78
9	0.83

Trace or redraw the faint graph below on paper. Keep the item ID visible on the page before photographing.



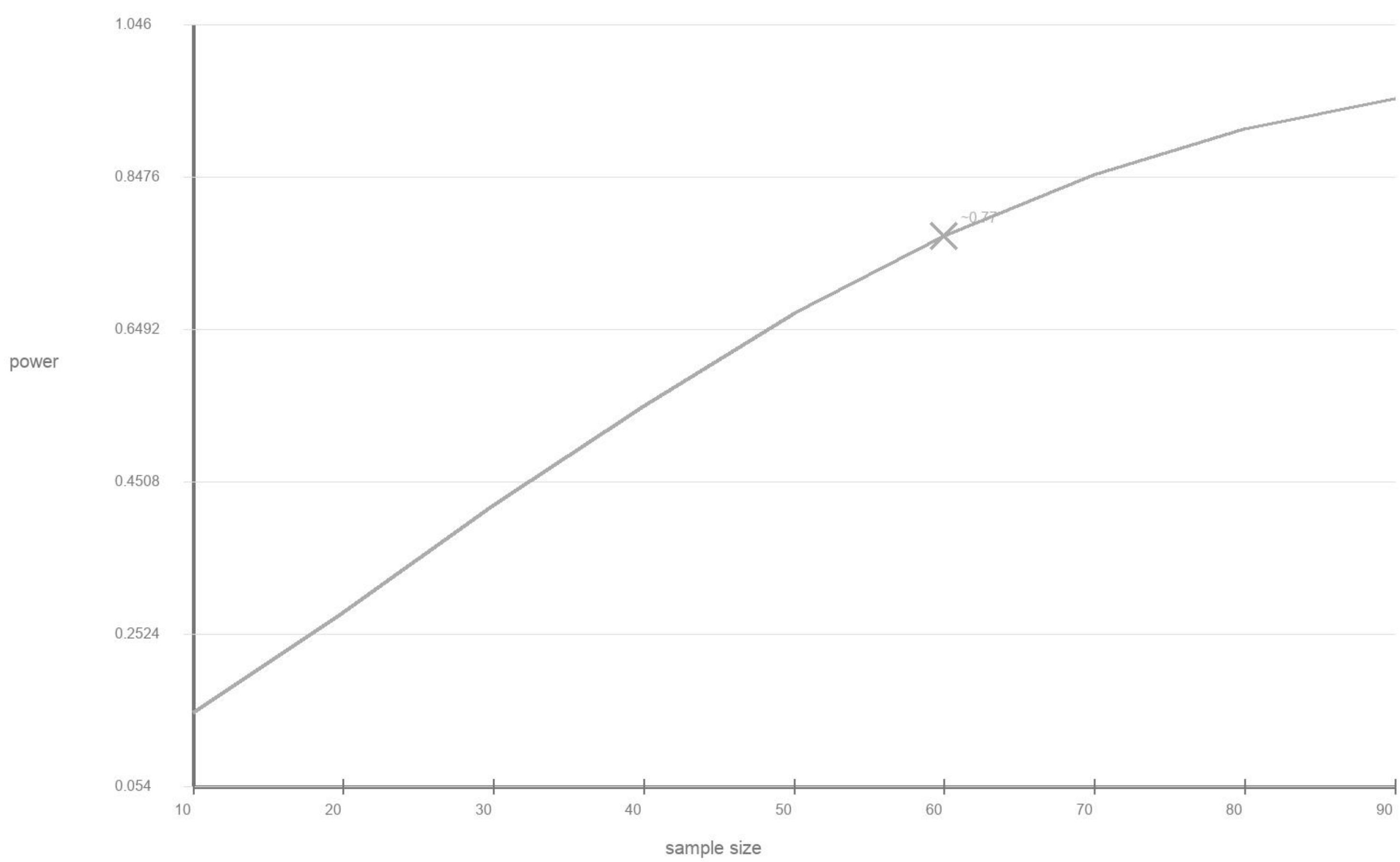
APSTATS-HDG-2026-GRAPH-039

The table gives values from a curve showing the power of a hypothesis test to detect a true effect as sample size increases. Draw the curve on a graph with sample size on the x-axis and power on the y-axis. Mark the point for sample size 60 with an X, then estimate the corresponding power.

Data table

Sample size	Power
10	0.15
20	0.28
30	0.42
40	0.55
50	0.67
60	0.77
70	0.85
80	0.91
90	0.95

Trace or redraw the faint graph below on paper. Keep the item ID visible on the page before photographing.



APSTATS-HDG-2026-GRAPH-040

The table gives values from a curve showing the proportion of a population that has adopted a new technology as months since launch increase. Draw the curve on a graph with months on the x-axis and adoption proportion on the y-axis. Mark the point for month 8 with an X, then estimate the corresponding adoption proportion.

Data table

Month	Adoption proportion
2	0.05
4	0.14
6	0.27
8	0.44
10	0.6
12	0.73
14	0.83
16	0.9
18	0.95

Trace or redraw the faint graph below on paper. Keep the item ID visible on the page before photographing.